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Salesforce CRM Development Track

A CRM Application for The Chai Shop & Co

A Salesforce-Based Customer Relationship Management System

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A SmartBridge Product

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User Story

As a sales representative at The Chai Shop & Co, I need one dependable CRM workspace where I can look after our customers and keep every sale on track.

- I want to open and update employer details quickly, and to tell online orders apart from walk-in orders through dedicated record types, so I can serve each customer in the way that suits them.
- I want the system to work out order quantity and the total bill by itself, so that every transaction is accurate and I never have to do the arithmetic by hand.
- I want ready-made reports and dashboards that turn raw sales figures into clear pictures, so that I and the management team can make better decisions and plan our strategy.
- I want the application to send each customer a birthday email with a special offer, so that our customers feel remembered and keep coming back.

1. Introduction

1.1 Project Overview

The project "A CRM Application for The Chai Shop & Co" develops a Salesforce-based Customer Relationship Management (CRM) system that manages customer, employer, and enterprise accounts efficiently. The system automates sales processes, maintains accurate customer records, generates reports and dashboards, and improves customer engagement through automated notifications.

The user story and project objectives are defined at the beginning of this document.

1.2 Purpose

The purpose of this project is to:

- Streamline customer relationship management
- Automate billing and order calculations
- Improve data accuracy
- Enable data-driven decision-making
- Enhance customer engagement through email automation

2. Ideation Phase

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Sales Representative	Record customer orders and prepare accurate bills	The totals I produce frequently turn out wrong	Quantities and bill amounts have to be worked out by hand	Under pressure and unproductive
PS-2	Store Manager	Keep track of sales performance and customer information	The reports I receive arrive late and leave gaps	Information is scattered across several separate files	Uncertain and unhappy with the process
PS-3	Customer	Receive offers and updates that feel personal to me	Messages from the shop rarely reach me on time	The shop has no automated way of contacting customers	Overlooked and unappreciated

2.2 Empathy Map Canvas

Target Users

- Sales Representatives
- Managers
- System Administrators
- Business Analysts

Empathy Analysis

Aspect	Description
Think & Feel	Need accurate customer data and reports
Hear	Feedback from management
See	Multiple registers and spreadsheets
Say & Do	Enter data manually
Pain	Errors, delays, data loss
Gain	Automation, fast reports

2.3 Brainstorming

Brainstormed Ideas

- Manual CRM system
- Excel-based management
- Web-based CRM
- Salesforce-based CRM
- Automated email system

Selected Idea

A Salesforce-based CRM Application for The Chai Shop & Co, combined with an automated email system for customer communication.

3. Requirement Analysis Phase

3.1 Customer Journey Map

Journey Stages

Stage	Activity	Outcome
1. Login	The user signs in to Salesforce with valid credentials	Secure access to the org
2. Access CRM Application	The user opens The Chai Shop & Co Lightning App	All CRM objects are available in one workspace
3. Enter Customer Data	Customer, employer, and enterprise details are captured	Complete, validated records
4. Generate Orders	Order items are selected for online or walk-in customers	Orders are stored under the correct record type
5. Calculate Bills	Formula fields compute order quantity and total bill	Accurate billing with no manual effort
6. View Reports	Sales and customer reports are run	Clear, current sales information
7. Decision Making	Managers review dashboards	Informed business decisions

3.2 Solution Requirements

Functional Requirements

FR No	Requirement	Description
FR-1	User Login	Authorized users sign in securely to the Salesforce org
FR-2	Object Creation	Employer, Customer, and Enterprise custom objects store business data
FR-3	Data Entry	Users enter and update records through page layouts and record types
FR-4	Billing Automation	Formula fields calculate order quantity and total bill
FR-5	Report Generation	Reports and dashboards summarize sales and customer data
FR-6	Email Automation	Flows send birthday offer emails to customers automatically

Non-Functional Requirements

NFR-1: Usability

The system should be easy to use for sales representatives, managers, administrators, and analysts without requiring technical training.

Requirements:

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-understand forms

- User-friendly dashboards

Example: A new sales employee should be able to log in, enter customer details, and generate orders within a few minutes without help. The Salesforce Lightning UI provides drag-and-drop features and easy navigation.

NFR-2: Security

The system must protect customer and business data from unauthorized access.

Requirements:

- Role-based access control
- Secure login authentication
- Encrypted data storage
- Data backup and recovery

Examples:

- Profiles and permission sets
- Object-level and field-level security
- Restricted billing access
- Secure login and password policies

NFR-3: Reliability

The CRM system should work correctly at all times and store data accurately without loss.

Requirements:

- Automatic data saving
- Regular system backups
- Error handling mechanisms
- Stable cloud environment

Example: If a user enters order data and the internet disconnects, the system should save the information and restore it later. The Salesforce cloud provides automatic backups.

NFR-4: Performance

The system should respond quickly and handle large amounts of data efficiently.

Requirements:

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal delay

Example: When a manager generates monthly sales reports for hundreds of customers, the report should load within 2–3 seconds. Optimized queries and caching improve speed.

NFR-5: Availability

The system should be accessible whenever required.

Requirements:

- 24/7 system availability
- Minimal downtime
- Cloud-based hosting
- Disaster recovery

Example: Managers should be able to access the CRM system during peak business hours, weekends, or late nights. The Salesforce cloud provides high uptime.

NFR-6: Scalability

The system should support business growth without affecting performance.

Requirements:

- Ability to handle more users
- Support for increasing customer data
- Expandable storage
- Consistent system speed

Example: If The Chai Shop & Co expands to multiple branches, the CRM system should support thousands of customers without slowing down. The Salesforce infrastructure scales automatically.

3.3 Data Flow Diagram

The data flows through the system in the following sequence:

User → Salesforce UI → Custom Objects → Validation Rules → Flows → Reports → Dashboards

Stage	Role in the Data Flow
User	Enters and retrieves data through the Lightning interface
Salesforce UI	Presents forms, tabs, and page layouts
Custom Objects	Store Employer, Customer, and Enterprise records (read/update)
Validation Rules	Check the logic and format of entered data
Flows	Automate tasks such as birthday email delivery
Reports	Summarize stored data
Dashboards	Present report results visually for decision-making

3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Database	Salesforce Cloud
Automation	Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

4. Project Design Phase

4.1 Problem-Solution Fit

The CRM system replaces manual records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely business insights.

Problem	Solution
Manual records are prone to errors, delays, and data loss	Cloud storage prevents data loss
Analysis is slow and unreliable	Dashboards improve analysis
Reports arrive late	Real-time reports eliminate delays

4.2 Proposed Solution

Parameter	Description
User Login & Authentication	Allows authorized users to securely log in to the CRM system using valid credentials.
Customer Record Management	Enables users to create, view, update, and delete customer details such as name, contact number, and address.
Employer & Enterprise Management	Allows management of employer and enterprise account information for business operations.
Order Management	Supports creation and tracking of customer orders, including order quantity and product details.
Automated Billing System	Automatically calculates the total bill amount using formula fields based on order quantity and pricing.
Validation Rules	Ensures accurate data entry by validating fields such as phone number, PAN, and Aadhaar.
Record Types & Page Layouts	Provides different record types and layouts for various customer and enterprise categories.
Global Picklists	Maintains standardized values for fields such as product type, category, and payment mode.
Email Notification System	Sends automated emails to customers for birthdays, offers, and order confirmations.
Workflow & Flow Automation	Automates tasks such as updating records and triggering alerts using Salesforce Flows.
Report Generation	Generates detailed reports on sales, customers, and enterprise performance.
Dashboard Visualization	Displays real-time analytics through interactive dashboards and charts.
Data Backup & Recovery	Ensures secure storage and recovery of business data in the Salesforce cloud.
User Role Management	Controls system access using profiles and permission sets for different users.

4.3 Solution Architecture

Architecture Layers

1. **Presentation Layer:** Lightning UI
2. **Application Layer:** Flows and Logic
3. **Data Layer:** Custom Objects
4. **Analytics Layer:** Reports and Dashboards

Design Note

This layered architecture ensures modularity, scalability, and maintainability.

5. Project Planning & Scheduling

5.1 Project Planning

The project followed an Agile-based development approach with defined milestones and sprints. Planning focused on building a structured Salesforce environment, modeling business data, automating routine activities, and delivering useful reports and dashboards within the planned timeline.

#	Planning Area	Planned Activity
01	Development Setup	Create the Salesforce Developer environment and prepare the project workspace.
02	Data Modeling	Design Customer, Employer and Enterprise objects, fields, relationships and tabs.
03	Configuration	Configure page layouts, record types, picklists and validation rules.
04	Automation	Develop email templates, flows and automated billing/notification processes.
05	Analytics	Build reports and dashboards for sales, customer and enterprise monitoring.
06	Sprint Review	Test each increment, review progress, resolve issues and track completion.

Key planning activities:

- A Salesforce Developer account was created to set up the development environment.
- Custom objects were designed to store customer, employer, and enterprise data.
- Custom tabs and a Lightning Application were developed for easy system navigation.
- Fields, relationships, and global picklists were configured to ensure data consistency.
- Page layouts and record types were created based on user roles and customer categories.
- Validation rules were implemented to prevent incorrect data entry.
- Email templates and automated flows were developed for customer notifications.
- Reports and dashboards were created for performance monitoring and analysis.
- Regular sprint reviews and testing were conducted to track progress and resolve issues.
- The project was completed within the planned timeline through systematic scheduling.

5.2 Planning Principles

- **Incremental delivery:** Functionality was organized into manageable sprints.
- **Data consistency:** Fields, relationships, picklists and validation rules were planned before automation.
- **User-focused design:** Layouts, record types and navigation were aligned with different CRM users.
- **Continuous review:** Sprint reviews and testing were used to identify and resolve issues early.

Planned Outcome

A scalable Salesforce CRM environment capable of managing customer, employer and enterprise information while supporting orders, billing, automated communication, reporting and dashboard-based analysis.

5.3 Product Backlog & Sprint Schedule

The backlog translates the project requirements into user stories. Story points indicate relative effort, while priority identifies the order in which work should be addressed.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the The Chai Shop & Co CRM application.	1	High	Moolai Riza Fathima
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Customer, Employer, and Enterprise objects so that business data can be stored and managed systematically.	3	High	Bushra Umme Kulsoom
Sprint-2	Data Modeling	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage CRM records within the application.	3	High	Labina Mariyam .K
Sprint-2	Data Modeling	USN-4	As a manager, I want a dedicated Lightning App so that all CRM operations are accessible from a single workspace.	3	High	Moolai Riza Fathima
Sprint-2	Data Modeling	USN-5	As an admin, I want to create customer and order fields so that complete sales and billing details can be captured accurately.	5	Medium	Moolai Aleena Maryam
Sprint-2	Data Modeling	USN-6	As a user, I want customized page layouts for different record types so that only relevant information is displayed during data entry.	5	High	Shahiya Saliqia .A
Sprint-2	Data Modeling	USN-7	As a manager, I want separate record types for Retail and Enterprise customers so that different business categories can be managed efficiently.	5	Medium	Bushra Umme Kulsoom
Sprint-3	Automation	USN-8	As a system, I want to validate customer details and order inputs so that incorrect data entries are prevented.	3	High	Labina Mariyam .K
Sprint-3	Automation	USN-9	As a system, I want automated flows to send email notifications so that customers receive timely updates and offers.	5	High	Moolai Aleena Maryam
Sprint-4	Automation	USN-10	As a system, I want to automatically calculate billing amounts so that invoices are generated without manual effort.	5	High	Shahiya Saliqia .A
Sprint-5	Reports	USN-11	As a manager, I want sales and customer reports so that I can monitor business performance and make informed decisions.	4	High	Bushra Umme Kulsoom
Sprint-6	Dashboards	USN-12	As a manager, I want visual dashboards so that I can quickly analyze sales, customers, and enterprise performance.	4	Medium	Moolai Riza Fathima

Workload distribution

Team Member	Role	User Stories Owned
Moolai Riza Fathima	Team Leader	USN-1, USN-4, USN-12 (environment setup, application integration, and final dashboard review)
Bushra Umme Kulsoom	Team Member	USN-2, USN-7, USN-11
Labina Mariyam .K	Team Member	USN-3, USN-8
Moolai Aleena Maryam	Team Member	USN-5, USN-9
Shahiya Saliqua .A	Team Member	USN-6, USN-10

Backlog Structure

The backlog moves from environment setup and data modeling toward automation and analytics. This sequence reduces dependency issues, because reporting and dashboards are built only after the underlying CRM data and processes are established.

5.4 Project Tracker & Sprint Control

The sprint tracker gives a compact view of planned effort, timing and completion. The original schedule defines six sprints in the backlog, while the tracker below records the first four sprint windows supplied in the project plan.

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 Oct 2026	06 Oct 2026	20	06 Oct 2026
Sprint-2	20	6 Days	07 Oct 2026	12 Oct 2026	—	—
Sprint-3	20	6 Days	13 Oct 2026	18 Oct 2026	—	—
Sprint-4	20	6 Days	19 Oct 2026	24 Oct 2026	—	—

5.5 Sprint Lifecycle

Plan	Build	Test	Review	Release
Select stories and estimate effort	Configure and develop	Validate behaviour and data	Review progress and issues	Deliver the completed increment

5.6 Estimation & Velocity + Burndown Tracking

Estimation & Velocity

Story points provide a relative measure of work rather than a direct time estimate. As sprints progress, completed story points can be used to observe team velocity and improve future planning.

- Each planned sprint carries a target of 20 story points over a 6-day window.
- Sprint-1 delivered all 20 planned points by its planned end date, which gives a velocity of 20 points per sprint (about 3.3 points per day).

- This velocity serves as the baseline for planning later sprints.

Burndown Tracking

A burndown chart plots the remaining story points against the sprint timeline. A downward trend toward zero indicates progress toward the planned sprint scope. Comparing the actual line with the ideal line at each sprint review helps the team spot delays early and rebalance work among members.

5.7 Planning Summary + Key Deliverables

From Requirements to Delivery

The planning phase converts CRM requirements into an organized development roadmap. It establishes what needs to be built, who owns each user story, how much effort is expected, and when each increment is scheduled.

Area	Summary
Requirements	Functional needs are converted into user stories and measurable tasks.
Schedule	Work is grouped into sprints with planned start and end dates.
Ownership	Each story is assigned to a team member for accountability.
Quality	Testing and sprint reviews are included throughout development.

Key Deliverables

- Salesforce development environment
- Customer, Employer and Enterprise data model
- Custom tabs, Lightning App, fields and relationships
- Page layouts, record types and global picklists
- Validation rules, email templates and automated flows
- Reports and dashboards for CRM analysis

Planning Phase Complete

The project plan establishes a clear path from setup and data modeling to automation, reporting and dashboard delivery.

6. Project Development Phase

The development phase converts the planned CRM design into a working Salesforce application. Implementation is organized as a sequence of milestones, beginning with the developer environment and progressing through objects, navigation, configuration, automation, reporting and dashboards.

6.1 Project Flow

Milestone	Stage	Purpose
01	Developer Account	Create and activate the Salesforce development environment.
02	Object Creation	Create Employer, Customer and Enterprise custom objects.
03	Tabs	Create custom tabs for easy navigation.
04	Lightning App	Bring CRM operations together in a dedicated application.
05	Global Picklist	Standardize repeated values across CRM fields.
06	Fields & Relationships	Define the information captured and how records connect.
07	Page Layouts	Control which fields and sections users see.
08	Record Types	Support different customer or enterprise categories.
09	Validation Rules	Prevent inaccurate or incomplete data entry.
10	Email Templates	Prepare reusable customer communication templates.
11	Flows	Automate notifications, updates and business processes.
12	Reports & Dashboards	Turn CRM data into monitoring and analysis tools.
13	Conclusion	Review the completed CRM solution and outcomes.

6.2 What You'll Learn

1. Real-time Salesforce project workflow
2. Standard and custom objects
3. Picklist value sets
4. Types of fields and relationships
5. Page layouts
6. Record types
7. Validation rules
8. Email templates
9. Flows
10. Reports and dashboards

6.3 Implementation: Milestone Summary

Milestone 1: Developer Account Creation

- Signed up for a free Salesforce Developer Edition org using first and last name, email, Role as Developer, college/company name, country (India), postal code, and a unique username in email format.
- Activated the account through the verification email (which can take 5–10 minutes to arrive), then set a password and security question.
- The user was redirected to Salesforce Setup, which confirmed that the development environment was ready.

Milestone 2: Objects (Employer, Customer, Enterprise)

- Opened Setup → Object Manager → Create → Custom Object for each of the three objects.
- Created **Employer Account**, **Customer Account** and **Enterprise Account**, each with a Text-type record name (e.g. "Employer Account Name").
- Enabled **Allow Reports**, **Allow Activities** and **Track Field History** on every object, and turned on **Allow Search**.
- Outcome: three data containers for employer, customer and enterprise information.

Milestone 3: Tabs

- Opened Setup → Tabs and selected **New** under Custom Object Tabs.
- Created a custom tab for Employer Accounts, Customer Accounts and Enterprise Accounts, choosing a tab style for each.
- Kept the default profile visibility and the option to append tabs to users' existing personal customizations.
- Outcome: direct navigation points to all three objects.

Milestone 4: The Lightning App

- Created a new Lightning App named **The Chai Shop & Co** from App Manager.
- Selected Standard navigation, the Setup experience, and Desktop and Phone form factors, with the default utility bar alignment.
- Added Customer Accounts, Employer Accounts, Enterprise Accounts, Reports and Dashboards as navigation items.
- Granted access to the **System Administrator** profile.
- Outcome: a single workspace for all CRM operations.

Milestone 5: Global Picklist

- Created four global picklist value sets from Setup → Picklist Value Sets.
- **Gender**: Male, Female, Others.
- **Items**: 17 products, covering tea powders, Osmania and other biscuits, cookies, and gift boxes (e.g. Platinum Tea Powder, Ragi Biscuits, Butter Almond Cookies, Signature Box).

- **Payment Mode:** Gpay, PhonePe, Paytm, Credit or Debit card, Cash.
- **Type of Items:** Tea powder, Cookies, Gift Hampers.
- Outcome: standardized, reusable values shared across objects.

Milestone 6: Fields & Relationships

- **Employer Account fields:** First Name, Last Name, Gender (global picklist), Phone Number, Email ID, Date of Birth, Aadhar Number (12 characters), PAN Details (10 characters), Designation (picklist: Manager, Finance and Accounting, Kitchen Staff, Cashier, Servers) and Salary (currency).
- **Customer Account fields:** First Name, Last Name, Gender, Phone Number, Date of Birth, Email ID, Address, Type of Items and Products Purchased (multi-select picklists from global sets), Chai (multi-select picklist of seven tea varieties), and Payment Mode.
- **Relationship:** a Lookup field, Associated Employer, links Customer and Enterprise records to Employer Account.
- **Formula fields:**
 - Date of Birthday is today? (Checkbox) drives the birthday automation.
 - Order Quantity (Number) counts the items selected.
 - Total Bill (Number) sums the price of each selected item.
- **Enterprise Account fields:** Enterprise Name, Phone Number, Email ID, Address, Type of Items, Products Purchased, Associated Employer (lookup), Payment Mode, plus Order Quantity and Total Bill formulas without the Chai terms.

Birthday check formula (Customer Account):

```
IF(
  AND(
    DAY( TODAY() ) = DAY( Date_of_Birth__c ),
    MONTH( TODAY() ) = MONTH( Date_of_Birth__c )
  ),
  TRUE,
  FALSE
)
```

Order Quantity formula (Customer Account):

```

IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Super Dust Tea Powder"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Choco chip Cookies"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Honey Oats Cookies"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Assorted Box"), 1, 0 ) +
IF(INCLUDES( Chai__c , "The Chai Shop & Co Special Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Single Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Masala Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Lemon Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Ginger Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Green Tea"), 1, 0 ) +
IF(INCLUDES( Chai__c , "Sugar Free Tea"), 1, 0 )

```

Total Bill formula (Customer Account):

```

IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 175, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 150, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Super Dust Tea Powder"), 125, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 50, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 80, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 160, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 170, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 190, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Choco chip Cookies"), 200, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 150, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Honey Oats Cookies"), 160, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 850, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1500, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Assorted Box"), 600, 0 ) +
IF(INCLUDES( Chai__c , "The Chai Shop & Co Special Tea"), 70, 0 ) +
IF(INCLUDES( Chai__c , "Single Tea"), 50, 0 ) +
IF(INCLUDES( Chai__c , "Masala Tea"), 60, 0 ) +
IF(INCLUDES( Chai__c , "Lemon Tea"), 40, 0 ) +
IF(INCLUDES( Chai__c , "Ginger Tea"), 55, 0 ) +
IF(INCLUDES( Chai__c , "Green Tea"), 45, 0 ) +
IF(INCLUDES( Chai__c , "Sugar Free Tea"), 55, 0 )

```

Milestone 7: Page Layouts (Online Orders, Walk-In Orders)

- Created two page layouts on the Customer Account object from Object Manager → Page Layouts → New.
- **Online Orders layout:** organized into Customer Details, Order Details and Employer Details sections, with Order Quantity and Total Bill shown in Order Details.
- **Walk-In Orders layout:** uses the same sections, and additionally shows the Chai selection and Payment Mode for in-store sales.

- Outcome: users see only the fields relevant to each order channel during data entry.

Milestone 8: Record Types (Online Order, Walk-In Order)

- Created two active record types on Customer Account, both based on the Master record type.
- **Online Order** is assigned the Online Orders page layout for all profiles.
- **Walk-In Order** is assigned the Walk-In Orders page layout for all profiles.
- Outcome: customer records are separated by order channel, which also allows reports to group sales by record type.

Milestone 9: Validation Rules (Aadhar, PAN, Phone on Employer / Customer / Enterprise)

- **Employer Account:** created three rules for Aadhar Number, PAN Number and Phone Number, each displaying its error at the top of the page.
 - The Aadhar rule requires exactly 12 numeric digits that never start with 0.
 - The PAN rule requires a non-blank value in the standard 10-character PAN format: five capital letters, four digits, and one capital letter (for example, ABCDE1234F).
 - The Phone rule requires exactly 10 numeric digits with no spaces, dashes or brackets.
- **Customer Account and Enterprise Account:** created a Phone Number validation rule on each, using the same 10-digit logic.
- Outcome: incorrect or incomplete identity and contact data is rejected at the time of entry.

Aadhar Number rule (Employer Account):

```
AND(
  NOT(ISBLANK( Aadhar_Number__c )),
  LEN( Aadhar_Number__c ) <> 12,
  NOT(REGEX( Aadhar_Number__c , "^[1-9]{12}$"))
)
```

PAN Number rule (Employer Account, field PAN_Details__c):

```
OR(
  ISBLANK( PAN_Details__c ),
  NOT(REGEX( PAN_Details__c , "^[A-Z]{5}[0-9]{4}[A-Z]{1}$"))
)
```

Phone Number rule (Employer, Customer and Enterprise Accounts):

```
AND(
  NOT(ISBLANK( Phone_Number__c )),
  NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
  NOT(REGEX( Phone_Number__c , "^\d+$"))
)
```

Milestone 10: Email Template + Email Alert (Birthday Offer)

- Created a **Text** Classic Email Template named **Birthday Offer Email** in the Unfiled Public Classic Email Templates folder, marked available for use.
- The subject line is "Special Birthday Offer Inside!". The body greets the customer by first name through the merge field `{!Customer_Account__c.First_Name__c}`.

- The email offers **50% off** the next purchase, redeemable with the code **HBDAY50**, and is signed off by The Chai Shop & Co.
- Created an **Email Alert** named Happy Birthday email alert on Customer Account using this template. The recipient is the customer's Email ID field and the sender is the default workflow user's email address.
- Confirmed that a Default Workflow User is assigned under Process Automation Settings so that the alert can be sent.

Milestone 11: Flows (Schedule-Triggered Birthday Flow)

- Created a **Schedule-Triggered Flow** in Flow Builder with a start date and time, running **Daily**.
- Set the object to **Customer Account** with all conditions required:
 - `Date_of_Birth_is_today__c` Equals True
 - `Email_ID__c` Is Null equals False (the customer must have an email)
- Added an **Action** element that calls the Happy Birthday email alert, labelled "Send Happy Birthday Email", with Record ID set to `{!$Record.Id}`.
- Saved the flow as **Happy Birthday Email Flow** and activated it.
- Outcome: every day, customers whose birthday falls on that date automatically receive the birthday offer email.

Milestone 12: Reports & Dashboards (Customer Sales, Enterprise Sales, Dashboard)

- **Customer Account Sales Report:** built from the Customer Accounts report type. It shows Type of Items, Products Purchased, Chai, Payment Mode, Order Quantity and Total Bill, grouped by Record Type (Online Order and Walk-In Order), with a chart added.
- **Enterprise Account Sales Report:** built from the Enterprise Accounts report type. It shows Enterprise Name, Products Purchased, Payment Mode, Order Quantity and Total Bill, grouped by Type of Items, with a chart added.
- **Dashboard:** created a dashboard named **Customer Vs Enterprise Sales** with two chart widgets, one per report, and saved it.
- Outcome: management can compare customer and enterprise sales at a glance.

Milestone 13: Conclusion

- The CRM application brings employer, customer (online and walk-in) and enterprise data into one system.
- Automated order quantity and total bill calculations reduce errors, so staff can concentrate on service.
- Automated birthday emails create personal communication and improve customer engagement.
- Reports and dashboards support informed decisions, which completes the planned objectives of the project.

6.4 CRM Implementation Architecture

Presentation	Application	Data	Analytics
Lightning UI	Flows & Logic	Custom Objects	Reports & Dashboards

6.5 Development Outcome

Development Outcome

The completed implementation brings together structured CRM data, controlled user access, consistent data entry, automated workflows, customer communication, and real-time business reporting within Salesforce.

6.6 Final Milestone Checklist

- ☒ Developer environment
- ☒ Custom objects
- ☒ Tabs & Lightning App
- ☒ Fields & relationships
- ☒ Page layouts & record types
- ☒ Validation rules
- ☒ Email templates & flows
- ☒ Reports & dashboards

7. Functional & Performance Testing

During the testing phase, screenshots were captured from the Salesforce Lightning interface to verify correct system behavior. The testing covered record creation, validation rules, record types and layouts, formula fields, flow execution, and reports and dashboards.

7.1 Functional Testing

Test ID	Feature Tested	Test Scenario	Expected Result	Result
TC-01	Record creation	Create Employer, Customer and Enterprise records with all required fields	Records save successfully and appear in the object tab and list view	Pass
TC-02	Required fields	Save a record with a required field (e.g. First Name) left blank	The system blocks the save and highlights the missing field	Pass
TC-03	Validation: Aadhar	Enter an Aadhar number that is not 12 digits, or contains non-numeric characters	Error message appears and the record is not saved	Pass
TC-04	Validation: PAN	Enter a PAN value that is blank or not in the format AAAAA9999A (five capital letters, four digits, one capital letter)	Error message appears and the record is not saved	Pass
TC-05	Validation: Phone	Enter a phone number that is not exactly 10 numeric digits	Error message appears and the record is not saved	Pass
TC-06	Validation: valid data	Enter a valid 10-digit phone number and a valid 12-digit Aadhar number	The record saves without any error	Pass
TC-07	Record types	Create a Customer Account using the Online Order record type, then the Walk-In Order record type	Each record is stored under the selected record type	Pass
TC-08	Page layouts	Open an Online Order record and a Walk-In Order record	Each record shows its assigned layout (Online Orders / Walk-In Orders)	Pass
TC-09	Formula: Order Quantity	Select several items in Products Purchased and Chai	Order Quantity equals the number of items selected	Pass
TC-10	Formula: Total Bill	Select items with known prices	Total Bill equals the sum of the selected items' prices	Pass
TC-11	Formula: Birthday check	Set a customer's Date of Birth to today's day and month	"Date of Birthday is today?" is checked	Pass
TC-12	Flow execution	Run the Happy Birthday Email Flow for a customer whose birthday is today and who has an email	The birthday offer email is sent to the customer's Email ID	Pass
TC-13	Customer sales report	Run the Customer Account Sales Report	Records are grouped by record type with quantity and bill totals	Pass
TC-14	Enterprise sales report	Run the Enterprise Account Sales Report	Records are grouped by Type of Items with quantity and bill totals	Pass
TC-15	Dashboard	Open the Customer Vs Enterprise Sales dashboard	Both widgets display chart data from the underlying reports	Pass

7.2 Performance Testing

Performance testing ensured fast response time and system stability during concurrent usage.

Parameter	Objective	Observation
Page load time	Record pages, tabs and list views open quickly	Pages loaded without noticeable delay
Report generation	Reports run within the 2–3 second target set in NFR-4	Reports and charts rendered promptly
Dashboard rendering	Dashboard widgets refresh smoothly	Widgets displayed without lag
Formula field calculation	Order Quantity and Total Bill update instantly on save	Values calculated immediately
Concurrent usage	Several users can work in the org at once	System remained stable
Automation reliability	The scheduled flow runs each day without failures	Flow executed as scheduled

8. Advantages & Disadvantages

8.1 Advantages

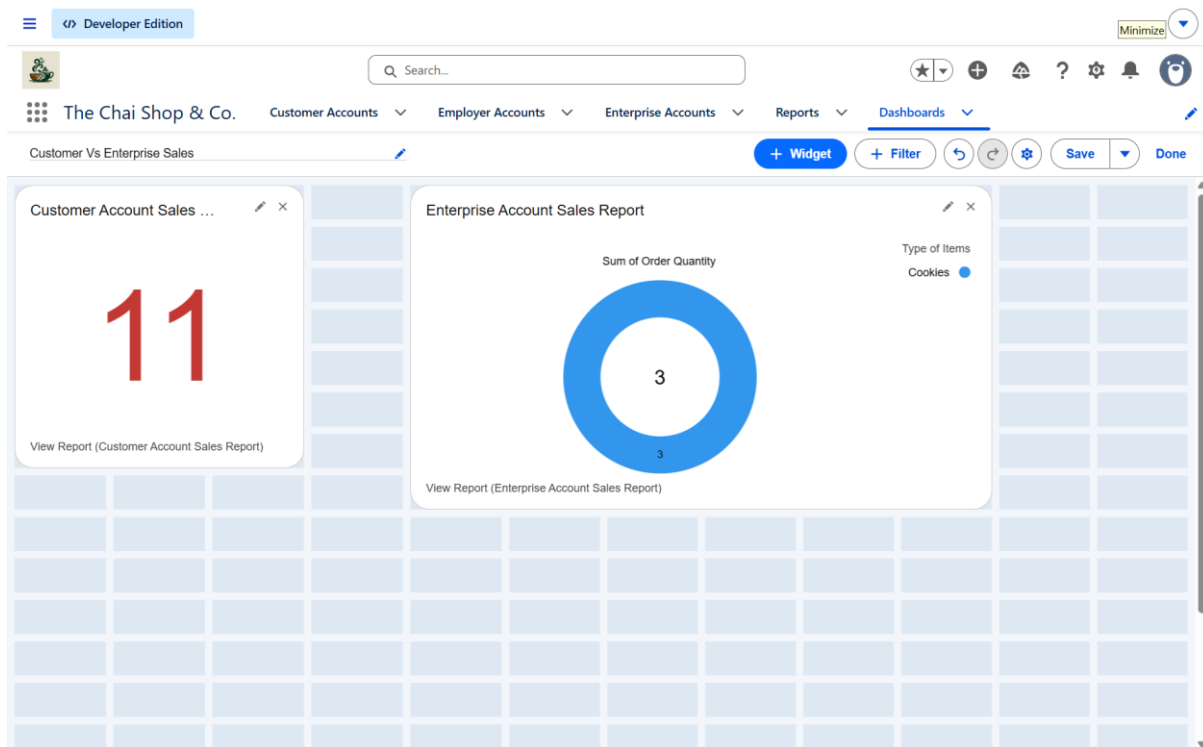
- Centralized data management
- Reduced manual work
- Automated calculations
- Improved customer engagement
- Real-time analytics

8.2 Disadvantages

- Requires a Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion

The CRM Application for The Chai Shop & Co successfully improves business operations by integrating customer data, automating billing, and enabling intelligent reporting. The system reduces errors, saves time, and enhances customer relationships through automation and analytics.



10. Future Scope

- Mobile CRM integration
- AI-based sales prediction
- WhatsApp/SMS alerts
- Inventory management
- Multi-branch support
- Advanced customer profiling

11. Appendix

- **Source Code:** Salesforce Configuration & Flows
- **Dataset:** Customer and Sales Records
- **Tools Used:** Salesforce Developer Org